A STACK ADMITTING AN ALGEBRAIC MORPHISM FROM AN ALGEBRAIC STACK IS ALGEBRAIC

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Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Base change of properties. algebraic-lemma-base-change-representable-transformations-property (cross-chapter, named). Transfers surjectivity and smoothness to the base-changed projection.
- Composition of properties. algebraic-lemma-composition-representable-transformations-property (cross-chapter, named). Composes the two smooth surjective legs.
- Smooth cover criterion. algebraic-lemma-smooth-surjective-morphism-implies-algebraic (cross-chapter, named). Concludes algebraicity.

Source reference. Original reference 045C: focused-seeds/geometry/S9-crit-algebraic-morphism-to-algebraic/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 045B: focused-seeds/geometry/S9-crit-algebraic-morphism-to-algebraic/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05UL: focused-seeds/geometry/S9-crit-algebraic-morphism-to-algebraic/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $F : \mathcal{X} \to \mathcal{Y}$ be a 1-morphism of stacks in groupoids over $(Sch/S)_{fppf}$. If*

(1) *$\mathcal{Y}$ is an algebraic stack, and*
(2) *F is algebraic (see above),*

then $\mathcal{X}$ is an algebraic stack.

Proof. By assumption (1) there exists a scheme T and an object ξ of $\mathcal{Y}$ over T such that the corresponding 1-morphism $\xi : (Sch/T)_{fppf} \to \mathcal{Y}$ is smooth an surjective. Then $\mathcal{U} = (Sch/T)_{fppf} \times_{\xi,\mathcal{Y}} \mathcal{X}$ is an algebraic stack by assumption (2). Choose a scheme U and a surjective smooth 1-morphism $(Sch/U)_{fppf} \to \mathcal{U}$. The projection $\mathcal{U} \longrightarrow \mathcal{X}$ is, as the base change of the morphism $\xi : (Sch/T)_{fppf} \to \mathcal{Y}$,

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surjective and smooth, see Algebraic Stacks, Lemma 045C. Then the composition $(Sch/U)_{fppf} \to \mathcal{U} \to \mathcal{X}$ is surjective and smooth as a composition of surjective and smooth morphisms, see Algebraic Stacks, Lemma 045B. Hence $\mathcal{X}$ is an algebraic stack by Algebraic Stacks, Lemma 05UL. □